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Assignment - 4

Section - A

1. What are the data structures that are used for clustering
Ans. Common data structures are K-d trees, R-trees, cluster feature trees (CF-trees), graphs & grids. It is used to store, organize and process data points efficiently for clustering algorithms.

2. Discuss various issues regarding classification.
Ans.
- Overfitting / Underfitting of model.
- Missing, noisy or incomplete data affecting accuracy

3. Write the formula for arithmetic distance used in K-Means algorithms.
Ans.
- Distance b/w two points $A(x_1, y_1)$ & $B(x_2, y_2)$:
  $$\text{Euclidean Dist.} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$
- Measures similarity based on geometric distance

4. What does BIRCH stand for?
Ans. BIRCH stands for Balanced Iterative Reducing and Clustering using Hierarchies. It is an efficient clustering technique for large datasets.

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5. What is dendrogram?

Ans. A tree-like hierarchical diagram representing cluste using Hierarchies. It is an efficient clustering technique for large datasets.

Section - B.

1. Write a short note on DBSCAN algo.

Ans. • DBSCAN stands for Density-Based Spatial Clustering of Applications with Noise.

• Groups data based on density connectivity of data points.

• Key Concepts:

-> ε (Epsilon): Neighbourhood radius.

-> MinPts: Minimum points req. to form a dense region

-> Core Points: Has ≥ MinPts within ε.

-> Border points: Less than MinPts but in a core point neighborhood.

-> Noise Point: Not belonging to any cluster:

• Advantages

-> Can detect clusters of arbitrary shape

-> Identifies noise & outliners.

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- Limitations :
→ Difficult to choose ε and MinPts.
→ Works poorly on varying densities.

2. What are data points and their types?

Ans. • Data Point : A set of values representing an object in N-dimensional space.

- Example : $(x_1, x_2, x_3 \ldots x_n)$
- Types of Data Points :

1. Core Points :
→ Have sufficient neighboring points (≥ MinPts)
→ Form the backbone of clusters.

2. Border Points :
→ Fewer points in their neighborhood but are within a core point's radius.

3. Noise / Outlier Points :
→ Do not belong to any cluster.
→ Far from dense regions.

- Important in density-based clustering like DBSCAN

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3. What is clustering and its types?

Ans: A technique of grouping similar data points into clusters such that intra-cluster similarity is high and inter-cluster similarity is low.

Types of clustering :-

1. Partitioning Clustering: Divides data into k clusters, eg: K-Means.
2. Hierarchical Clustering: Builds cluster tree using agglomerative/ divisive approach.
3. Density-Based Clustering: Forms clusters based on dense regions, eg: DBSCAN.
4. Grid-Based Clustering: Uses a grid to structure data space into cells, eg: STING.
5. Model-Based Clustering: Assumes a model for clusters, eg: EM algorithm.

Section - C

1. Explain K-Means algorithm with suitable eg.

Ans: K-Means is a partitioning clustering algorithm that divides data into k clusters based on distance. The objective is to minimize the within-cluster sum of square (WCSS)

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- Steps →

a) Choose a no. of cluster k.

b) Initialize k centroids randomly.

c) Assign each data point to nearest centroid using distance formula.

d) Recalculate centroids by taking mean of assigned points.

e) Repeat until centroids do not change.

- Example :-

→ Given data : (2,3), (3,3), (6,7), (7,8).

K = 2. → Cluster 1: near (3,3), Cluster 2: near (7,8)

After iterations, clusters stabilize into 2 groups.

- Advantages → Simple, fast, efficient, for large datasets.
- Disadvantages → Requires k in advance, sensitive to outliers.

Q. Differentiate b/w methods of hierarchical clustering.

Ans.
- Hierarchical clustering builds a tree-based representation (dendrogram).
- Two approaches:

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a) Agglomerative (Bottom-up):
→ Starts with each point as a seperate cluster.
→ Merges closest cluster step by step.
→ Methods include: single linkage, complete linkage, average linkage
→ Stops when only one cluster remains.

b) Divisive (Top-Down):
→ Starts with all points in one cluster.
→ Splits clusters recursively until each point is seperate.

• Differences:
→ Agglomerative: Merging process, Divisive: Splitting pro
→ Agglomerative is more commonly used.
→ Divisive is computationally more complex.

3 What do you mean by outlier analysis? Briefly describe types of outliers.

Ans. • Outlier Analysis:
→ Process of detecting data points that significantly differ from the majority.
→ Outliers may indicate fraud, errors, rare events or abnormalities.

- Types of Outliers:

a) Global (Point) Outliers:

→ Data point far from rest of dataset. Eg → Credit card fraud transactions.

b) Contextual Outliers:

→ Outlier based on context such as time / locations.

→ Eg: High temperature in winter.

c) Collective Outliers:

→ Group of data points collectively acting unusually.

→ Eg: Sudden spike in network traffic indicating attacks.

- Importance: Helps improve data quality, fraud detection, cyber security, health monitoring etc.